

# Sara Norouzinia

+1(404)644-1722 | U.S. Citizen | [sara.norouzinia@gmail.com](mailto:sara.norouzinia@gmail.com) | [linkedin.com/in/sara-norouzinia-32abc](https://www.linkedin.com/in/sara-norouzinia-32abc) | [saranorouzinia.org](https://saranorouzinia.org)

## Education

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| <b>Georgia Institute of Technology</b>                                                                                                                                                                               | Atlanta, GA |
| <i>Bachelor of Science in Computer Science (GPA: 3.74)</i>                                                                                                                                                           | May 2027    |
| <i>Master of Science in Computer Science, Specialization in Machine Learning</i>                                                                                                                                     | May 2028    |
| • <b>Relevant Coursework:</b> Object-Oriented Programming; Data Structures & Algorithms; Introduction to AI; Computer Organization and Programming; Systems and Networks; Design and Analysis of Algorithms; Game AI |             |

## Technical Skills

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| <b>Languages:</b> Python, Java, C/C++, Assembly, JavaScript, TypeScript, HTML, CSS, Bash                                                                             |
| <b>Frameworks &amp; Tools:</b> React, Node.js, Django, FastAPI, Docker, Jest UI, PyTorch, MongoDB, Jira, VSCode, Overleaf, Bootstrap, Git, GitHub, Figma, Linux/Unix |
| <b>Development Practices:</b> Agile, Scrum, CI/CD, Web Dev, System Design & Architecture, Machine Learning, NLP, CNN                                                 |

## Experience

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| • <b>Software Engineering Intern</b>                                                                                                                                                                          | May 2026 – August 2026 |
| <i>GTRI Applied Systems Lab</i>                                                                                                                                                                               | Huntsville, AL         |
| • Wrote 6 end-to-end automation test suites in Python and C++ to validate MongoDB request handling, surfacing read/write operation bugs before they reach production                                          |                        |
| • Resolved XML defects in a middleware layer between MongoDB to a web app, improving cross-service communication reliability                                                                                  |                        |
| • Refactored 5 React components for web application responsiveness and accessibility, extending Jest UI for testing                                                                                           |                        |
| • <b>Software Engineering Intern</b>                                                                                                                                                                          | May 2026 – August 2026 |
| <i>TechX AI Systems and LLM Engineering</i>                                                                                                                                                                   |                        |
| • Built a filter-based agent-lookup feature across 5 FastAPI endpoints, supporting sort/filter by popularity, relevance, & metadata                                                                           |                        |
| • Authored 8 backend API tests covering 4 endpoints and developed prompt templates, mock runners, and tracing instrumentation to streamline debugging                                                         |                        |
| • Implemented tool-selection and routing logic for an agentic AI system using MCP                                                                                                                             |                        |
| • <b>Machine Learning Researcher</b>                                                                                                                                                                          | Jan 2026 – May 2026    |
| <i>Anomaly Sound Detection in the Ocean Soundscape</i>                                                                                                                                                        | Metz, France           |
| • Accepted for <b>International Conference on Pattern Recognition</b> and published in Springer's LNCS series                                                                                                 |                        |
| • Programmed and trained a supervised ConvNeXt-based CNN in PyTorch for binary anomaly classification on a 600 audio dataset and validated against a 100-sample expert-labeled ground truth                   |                        |
| • Built a deep learning pipeline tuning via BCEWithLogitsLoss, early stopping, and ReduceLROnPlateau scheduling with L2-regularized Adam optimization to stabilize convergence on high-variance acoustic data |                        |
| • Evaluated model performance using F1-score, precision/recall metrics, and confusion matrix                                                                                                                  |                        |
| • <b>Machine Learning Engineer</b>                                                                                                                                                                            | Aug 2025 – Present     |
| <i>Data Driven Education – VIP Undergraduate Research</i>                                                                                                                                                     | Atlanta, GA            |
| • Collaborated in a cross-functional team to fine-tune a BART model for automated question-difficulty classification and generation on real student exam data                                                 |                        |
| • Built a linguistic feature-extraction pipeline using NumPy and textstat, computing readability metrics to score difficulty                                                                                  |                        |
| • Architected a dual-pipeline system (generation & classification) to cross-validate model outputs and improve prediction reliability                                                                         |                        |

## Projects & Research

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| • <b>Align Web App</b> – Backend Developer                                                                                        | Jan 2025 – May 2025 |
| • Built backend infrastructure using Django models and Google OAuth2, integrating Google Calendar to support 10+ app features     |                     |
| • Engineered event-driven triggers to automate system actions based on real-time user state changes, reducing manual coordination |                     |
| • Collaborated in a 5-person Agile team, coordinating schema and API design through GitHub version control and sprint planning    |                     |
| • <b>Personal Portfolio Website</b>                                                                                               | Jul 2025            |
| • Modeled and produced a fully responsive personal website using React, TypeScript, JavaScript, HTML, and CSS                     |                     |
| • Implemented reusable components to create a scalable, maintainable front-end application                                        |                     |
| • Prototyped end-to-end UI/UX flows and wireframes in Figma and Milanote prior to implementation                                  |                     |
| • <b>Vehicle Gas Consumption Analysis using IoT and GIS</b>                                                                       | Aug 2023 – May 2024 |
| • Designed a data-driven system integrating IoT data and GIS mapping to analyze vehicle fuel consumption patterns                 |                     |
| • Conducted a one-group pre-test/post-test experimental study across 10 random participants                                       |                     |
| • Built data pipelines to process, validate, and analyze large-scale sensor datasets for statistical analysis                     |                     |

## Leadership

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| • <b>GT Persian-American Resource Society – Executive Board</b>                                 |             |
| Planned 10+ events, facilitated member engagement, & coordinated with UGA and KSU clubs         | Atlanta, GA |
| • <b>Teaching Assistant – Object-Oriented Programming</b>                                       |             |
| Led 20+ student review sessions, supported individuals in office hours, and graded deliverables | Atlanta, GA |
| • <b>IBM SkillsBuild Student Ambassador</b>                                                     |             |
| Connected with future IBM skillbuilders and taught them about the importance of AI skills       | Atlanta, GA |
| • <b>Team Lead – Kaldi's Coffee</b>                                                             |             |
| Ran daily operations across 3 cafes, supported staff, and delegated team responsibilities       | Atlanta, GA |